

Experiment 03:

7. Laboratory Exercise

- Vigenere Cipher

Procedure:

1. Write a program in Python to encrypt and decrypt the given input using Vigenere cipher.
2. Test the output of program for following Inputs:
 - a. Plaintext = "She is Listening" with keyword: Pascal. Also check if decryption works.

Code:

```
def encrypt(text, key):
    cipher = ""
    key = key.upper()
    keyIndex = 0

    for character in text:
        if character == " ":
            cipher += " "
            continue

        plain = ord(character.upper()) - ord("A")
        keyCharacter = ord(key[keyIndex % len(key)]) - ord("A")

        encrypted = (plain + keyCharacter) % 26
        cipher += chr(encrypted + ord("A"))

        keyIndex += 1

    return cipher

def decrypt(cipher, key):
    text = ""
    key = key.upper()
    keyIndex = 0

    for character in cipher:
        if character == " ":
            text += " "
            continue

        c = ord(character.upper()) - ord('A')
        k = ord(key[keyIndex % len(key)]) - ord('A')
        p = (c - k) % 26

        text += chr(p + ord('A'))
        keyIndex += 1

    return text

plain = input("Enter Plain Text: ").replace(" ", "").upper()
key = input("Enter Key: ").upper()
```

```
blockSize = len(key)

plainBlocks = []
keyBlocks = []
cipherBlocks = []

for startIndex in range(0, len(plain), blockSize):
    plainBlock = plain[startIndex:startIndex + blockSize]

    # Match the key length to the current block
    keyBlock = key[:len(plainBlock)]

    cipherBlock = encrypt(plainBlock, key)

    plainBlocks.append(plainBlock)
    keyBlocks.append(keyBlock)
    cipherBlocks.append(cipherBlock)

print("\nPlain :", " ".join(plainBlocks))
print("Key   :", " ".join(keyBlocks))
print("Cipher:", " ".join(cipherBlocks))

cipherText = "".join(cipherBlocks)
decryptedText = decrypt(cipherText, key)

print("\nEncrypted Text:", cipherText)
print("Decrypted Text:", decryptedText)
```

Output:

```
Enter Plain Text: She is Listening
Enter Key: Pascal

Plain : SHEISL ISTENI NG
Key   : PASCAL PASCAL PA
Cipher: HHWKSX XSLGNT CG

Encrypted Text: HHWKSXSLGNTCG
Decrypted Text: SHEISLISTENING
```

b. Plaintext = “The house is being Sold Tonight” with keyword: Dollars

Output:

```
Enter Plain Text: The house is being Sold Tonight
Enter Key: Dollars

Plain : THEHOUS EISBEIN GSOLDTO NIGHT
Key   : DOLLARS DOLLARS DOLLARS DOLLA
Cipher: WVPSOLK HWDMEZF JGZWDKG QWRST

Encrypted Text: WVPSOLKHWDMEZFJGZWDKGQWRST
Decrypted Text: THEHOUSEISBEINGSGOLDTONIGHT
```

c. Plaintext = Your complete name (Name, middle name, surname) with keyword: Hello

```
Enter Plain Text: Shivam Shailesh Pimple
Enter Key: Hello

Plain : SHIVA MSHAI LESHPI MPLE
Key   : HELLO HELLO HELLO HELLO
Cipher: ZLTGO TWSLW SIDSD PQAWS

Encrypted Text: ZLTGOTWSLWSIDSDPQAWS
Decrypted Text: SHIVAMSHAI LESHPI MPLE
```

3. Test the output of program for following Inputs:

Code:

```
def decrypt(cipher, key):
    text = ""
    key = key.upper()
    keyIndex = 0

    for character in cipher:
        if character == " ":
            text += " "
            continue

        c = ord(character.upper()) - ord('A')
        k = ord(key[keyIndex % len(key)]) - ord('A')
        p = (c - k) % 26

        text += chr(p + ord('A'))
        keyIndex += 1

    return text

cipher = input("Enter Cipher Text: ").replace(" ", "").upper()
key = input("Enter Key: ").upper()

blockSize = len(key)

cipherBlocks = []
keyBlocks = []
plainBlocks = []

for startIndex in range(0, len(cipher), blockSize):
    cipherBlock = cipher[startIndex:startIndex + blockSize]

    keyBlock = key[:len(cipherBlock)]
    plainBlock = decrypt(cipherBlock, key)
    cipherBlocks.append(cipherBlock)
    keyBlocks.append(keyBlock)
    plainBlocks.append(plainBlock)

print("\nCipher :", " ".join(cipherBlocks))
print("Key      :", " ".join(keyBlocks))
print("Plain    :", " ".join(plainBlocks))

plainText = "".join(plainBlocks)
print("\nDecrypted Text:", plainText)
```

a. Ciphertext = “SMFPBZMYLWHMZYRAKPZIS ” with keyword: HEALTH

Output:

```
Enter Cipher Text: SMFPBZMYLWHMZYRAKPZIS
Enter Key: HEALTH

Cipher : SMFPBZ MYLWHM ZYRAKP ZIS
Key    : HEALTH HEALTH HEALTH HEA
Plain  : LIFEIS FULLOF SURPRI SES

Decrypted Text: LIFEISFULLOFSURPRISES
```

b. Ciphertext = “OINCMMBLSRKHJMVSIYIITW ” with keyword: security

4. Output:

```
Enter Cipher Text: OINCMMBLSRKHJMVSIYIITW
Enter Key: security

Cipher : OINCMMBL SRKHJMVSIYIITW
Key    : SECURITY SECURITY SECURIT
Plain  : WELIVEIN ANINSECU REWORLD

Decrypted Text: WELIVEINANINSECUREWORLD
```